



**Year 11 Preliminary Mathematics Advanced
Assessment Task 2
Term 3, Week 3, 2021**

Name: _____

Teacher: _____

Wednesday 28th July 2021

Setter: VN

- All answers are to be written on your own lined paper so they can be scanned and uploaded as a single PDF file to your teacher's google classroom. Make sure that you write your name on each of the lined pages used.
- **Start each question on a new page.**
- Board approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- For any multiple-choice question, **write the letter** that corresponds to your answer on your writing paper.
- Total possible mark: 32
- Time Allowed: **55 minutes inclusive of task downloading then answering on your own lined paper, scanning and uploading a single PDF file for your responses to your teacher's google classroom.**

Question 1 (5 marks)	Start a new page	Marks
(a) Which expression is equivalent to $\cot x + \tan x$?		1
A. 1 B. $\operatorname{cosec} x \sec x$ C. $\sec x + \operatorname{cosec} x$ D. 2		
(b) Differentiate $y = x^3 - 2x^2 - 5$ with respect to x .		1
(c) Find the exact value of $\sin 300^\circ$.		1
(d) Given $\sqrt[3]{a} = b^5$, make x the subject of the formula.		2

Question 2 (5 marks)**Marks**

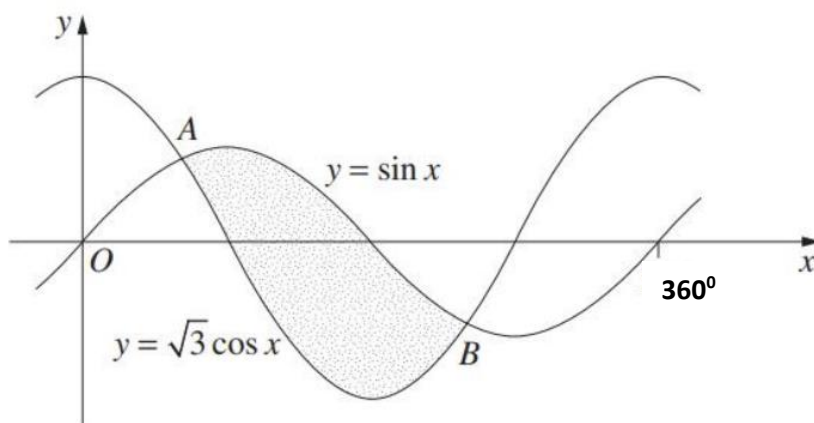
- (a) Which of the following is equal to $\frac{\log_2 9}{\log_2 3}$? 1

A. 2 B. 3 C. $\log_2 3$ D. $\log_2 6$

- (b) The graph below shows the curves $y = \sin x$ and $y = \sqrt{3} \cos x$. 2

The first two points of intersection to the right of the y axis are labelled **A** and **B**.

Solve the equation $\sin x = \sqrt{3} \cos x$ to find the x -coordinates of **A** and **B**.



- (c) Solve $\log_2 x + \log_2 (x+7) = 3$. 2

Question 3 (5 marks) [Start a new page](#)

- (a) What is the derivative of $\frac{2}{3x^4}$? 1

A. $-\frac{8}{3x^3}$ B. $-\frac{8}{3x^5}$ C. $\frac{8}{3x^3}$ D. $\frac{8}{3x^5}$

- (b) Differentiate with respect to x , leaving your answer fully factorised and in simplest positive integer or surd form, where appropriate.

(i) $y = (3x+2)(x-1)^3$ 2

(ii) $f(x) = \frac{x^3-3}{2x+5}$. 2

Question 4 (5 marks)**Start a new page****Marks**

(a) Which interval gives the range of the function $y = 5 + 2 \cos 3x$?

1

A. [2,8]

B. [3,7]

C. [4,6]

D. [5,9]

(b) Sketch the curve $y = \log_2(x+1)$ clearly showing any intercepts and asymptotes.

2

(c) Write $\frac{128}{2^n \times 64^{1-2n}}$ in the form 2^{a+bn} , where a and b are integers.

2**Question 5 (6 marks)****Start a new page**

(a) Given that $\frac{\sec x}{\operatorname{cosec} x} = \tan x$, solve, $3 \sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right)$ for $0 \leq x \leq 360^\circ$.

2

(b) Find the equation of the tangent to the parabola $y = 4x - x^2$ at the point where the gradient is -2 .

2

(c) For the function $f(x) = x^2 + 2$, show that $f'(x) = 2x$ by differentiating from first principles.

2

Question 6 (6 marks). Start a new page

Marks

- (a) Given that $x^2 + y^2 = 23xy$, express $\left(\frac{x+y}{5}\right)^2$ in terms of xy . **3**

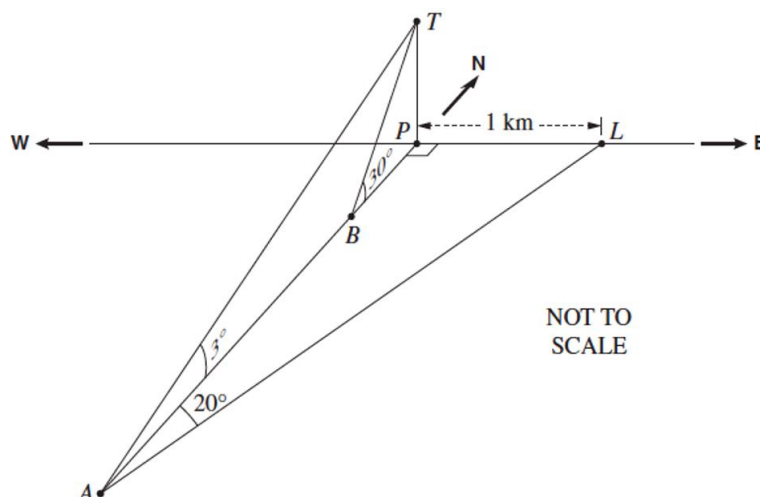
Hence or otherwise show that $\log\left(\frac{1}{5}(x+y)\right) = \frac{1}{2}(\log x + \log y)$.

- (b) A boat is sailing due north from a point **A** towards a point **P** on the shore line. The shore line runs from west to east.

In the diagram, **T** represents a tree on a cliff vertically above **P**, and **L** represents a landmark on the shore. The distance **PL** is **1km**.

From **A** the point **L** is on a bearing of 020° , and the angle of elevation to **T** is 3° .

After sailing for some time the boat reaches point **B**, from which the angle of elevation to **T** is 30° .



Show that $BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$ kilometres. **3**

End of Assessment



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Suggested Solution (s)	Comments
Question 1 a) $\cot x + \tan x$ $= \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}$ $= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$ $= \frac{1}{\sin x \cos x}$ $= \operatorname{cosec} x \sec x$ $\therefore \mathbf{B}$ (1 mark) b) $y' = 3x^2 - 4x$ (1 mark) c) $y = \sin 300^\circ$ $= -\sin 60^\circ$ $= -\frac{\sqrt{3}}{2}$ (1 mark) d) $\sqrt[x]{a} = b^5$ OR $\frac{1}{x} = \log_a b^5$ using the definition of a logarithm (1 mark) $a^{\frac{1}{x}} = b^5$ $x = \frac{1}{\log_a b^5}$ or $x = \frac{1}{5 \log_a b}$ (1 mark) $\log a^{\frac{1}{x}} = \log b^5$ (1 mark) $\frac{1}{x} \log a = 5 \log b$ $x = \frac{\log a}{5 \log b}$ (1 mark)	
Question 2 a) $\frac{\log_2 9}{\log_2 3}$ $= \frac{2 \log_2 3}{\log_2 3}$ $= 2 \therefore \mathbf{A}$ (1 mark)	



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Suggested Solution (s)	Comments
b) $\sin x = \sqrt{3} \cos x$ $\frac{\sin x}{\cos x} = \sqrt{3}$ $\tan x = \sqrt{3}$ (1 mark) $\therefore x = 60^\circ$ or $x = 240^\circ$ The x coordinates of the points A and B are $x = 60^\circ$ and $x = 240^\circ$ respectively. (1 mark) c) $\log_2 x + \log_2 (x+7) = 3$ $\log_2 x(x+7) = 3 \log_2 2$ $\log_2 x(x+7) = \log_2 2^3$ (1 mark) $x(x+7) = 2^3$ $x^2 + 7x - 8 = 0$ $(x+8)(x-1) = 0$ $x = -8$ or $x = 1$ But x cannot be negative (see step 1) hence $x = 1$ (1 mark) Question 3 a) $\frac{2}{3x^4}$ The derivative is: $\frac{2}{3} \times -4x^{-5} = -\frac{8}{3}x^{-5} = -\frac{8}{3x^5}$ The answer is B (1 mark) b) $y' = (x-1)^3 \frac{d}{dx}(3x+2) + (3x+2) \frac{d}{dx}((x-1)^3)$ $y' = 3(x-1)^3 + 3(x-1)^2(3x+2)$ (1 mark) $y' = 3(x-1)^2((x-1) + (3x+2))$ $y' = 3(x-1)^2(4x+1)$ (1 mark)	$x_A = \frac{\pi}{3}, x_B = \frac{4\pi}{3}$ acceptable



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c)

$$f'(x) = \frac{(2x+5) \times 3x^2 - 2(x^3-3)}{(2x+5)^2}$$

(1 mark)

$$f'(x) = \frac{6x^3 + 15x^2 - 2x^3 + 6}{(2x+5)^2}$$

$$f'(x) = \frac{4x^3 + 15x^2 + 6}{(2x+5)^2}$$

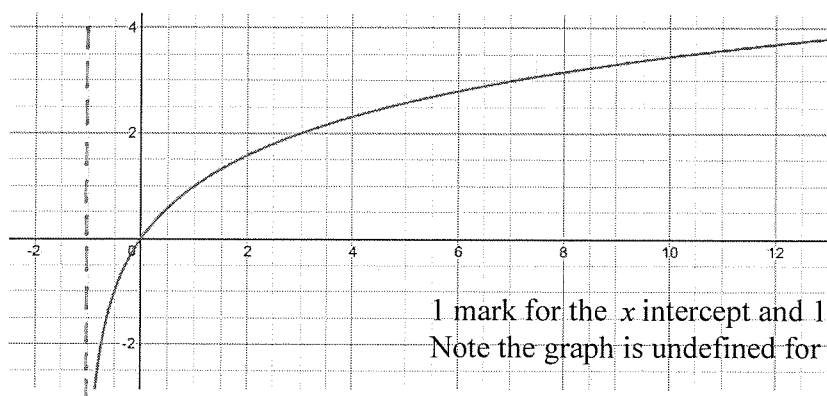
(1 mark)

Question 4

a) B

(1 mark)

b)



1 mark for the x intercept and 1 mark for the shape
Note the graph is undefined for $x \leq -1$

$$\text{c) } \frac{128}{2^n \times 64^{1-2n}}$$

$$= \frac{2^7}{2^n \times (2^6)^{1-2n}}$$

$$= \frac{2^7}{2^n \times 2^{6-12n}}$$

$$= \frac{2^7}{2^{6-11n}}$$

$$= 2^{7-(6-11n)}$$

$$= 2^{1+11n}$$

(1 mark)

(1 mark)



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Question 5

a) $3 \sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right)$ for $0 \leq \frac{x}{2} \leq 180^\circ$

$$\frac{\sec^2\left(\frac{x}{2}\right)}{\operatorname{cosec}^2\left(\frac{x}{2}\right)} = \frac{1}{3}$$

$$\left(\frac{\sec\left(\frac{x}{2}\right)}{\operatorname{cosec}\left(\frac{x}{2}\right)}\right)^2 = \frac{1}{3}$$

$$\frac{\sec\left(\frac{x}{2}\right)}{\operatorname{cosec}\left(\frac{x}{2}\right)} = \pm \frac{1}{\sqrt{3}}$$

(1 mark to get to this point)

$$\tan\left(\frac{x}{2}\right) = \pm \frac{1}{\sqrt{3}}$$

$$\therefore \frac{x}{2} = 30^\circ, 150^\circ$$

This means $\therefore x = 60^\circ$ or 300°

(1 mark for the correct answer)

Radian Measure acceptable and so

$$\therefore \frac{x}{2} = \frac{\pi}{6}, \frac{5\pi}{6}$$

This means $x = \frac{\pi}{3}$ or $\frac{5\pi}{3}$.

b) $y' = 4 - 2x$

when $y' = -2$

$$4 - 2x = -2$$

$$x = 3$$

When $x = 3$, $y = 3$

(1 mark for the correct x and y value)

$$y - 3 = -2(x - 3)$$

$y = -2x + 9$, which is the equation of the tangent. (1 mark for the equation of the

tangent)



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c)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2 - (x^2 + 2)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 2 - x^2 - 2}{h} \quad (1 \text{ mark for this step and 1 mark to get to the penultimate step})$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 2x + h$$

$$f'(x) = 2x$$

Question 6

a)

$$\left(\frac{x+y}{5}\right)^2 = \frac{x^2 + 2xy + y^2}{25}$$

$$\left(\frac{x+y}{5}\right)^2 = \frac{23xy + 2xy}{25}$$

$$\left(\frac{x+y}{5}\right)^2 = xy \quad (1 \text{ mark})$$

$$\therefore \log\left(\frac{1}{5}(x+y)\right)^2 = \log(xy) \quad (1 \text{ mark})$$

$$2\log\left(\frac{1}{5}(x+y)\right) = \log x + \log y$$

$$\log\left(\frac{1}{5}(x+y)\right) = \frac{1}{2}(\log x + \log y) \quad (1 \text{ mark})$$

b) In right triangle *APL*

$$\tan 20^\circ = \frac{1}{AP}$$

$$AP = \frac{1}{\tan 20^\circ} \quad (1 \text{ mark})$$



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In right triangle *APT*

$$\tan 3^\circ = \frac{TP}{AP}$$

$$\therefore \tan 3^\circ = TP \div \frac{1}{\tan 20^\circ}$$

$$\frac{\tan 3^\circ}{\tan 20^\circ} = TP$$

(1 mark)

Finally, in right triangle *BPT*

$$\tan 30^\circ = \frac{TP}{BP}$$

$$\therefore BP = \frac{TP}{\tan 30^\circ}$$

$$BP = \frac{\tan 3^\circ}{\tan 20^\circ} \div \frac{1}{\sqrt{3}}$$

$$BP = \frac{\sqrt{3} \tan 3^\circ}{\tan 20^\circ}$$

(1 mark)